

CHIBUIKEM PRECIOUS EZEMADUKA

Website: <https://chibuikemezmaduka.github.io/> | ezemadukachibuikem@gmail.com | Troy, NY 12180
[LinkedIn](#)

SUMMARY

Engineer and researcher working at the intersection of artificial intelligence and next-generation wireless networking, developing novel machine learning algorithms for spectrum management, dynamic spectrum sharing, wireless security, and 5G/6G/Wi-Fi connectivity. First-author publications in top IEEE venues, including IEEE Transactions on Cognitive Communications and Networking and IEEE Military Communications, plus industry experience at IBM Research, CableLabs, and AT&T Labs.

EDUCATION

Ph.D., Electrical Engineering

Expected Dec 2026

Department: Electrical, Computer and Systems Engineering
Rensselaer Polytechnic Institute, Troy, NY

Dissertation: Enhancing Spectrum Enforcement in Dynamic Spectrum Access Systems Using Deep Learning
Candidacy conferred: January 2025 • Advisor: Dr. Alhussein Abouzeid

M.S., Computer & Systems Engineering

May 2024

Department: Electrical, Computer and Systems Engineering
Rensselaer Polytechnic Institute, Troy, NY

Relevant Coursework: Deep Learning, Machine Learning from Data, Deep Learning on Graphs, Detection & Estimation Theory, Stochastic Signals & Systems, ML for Autonomous Systems, Computational Optimization, Security & Privacy of ML, Internetworking of Things

B.Eng., Electrical & Electronic Engineering (Communications Engineering)

Feb 2018

Federal University of Technology, Owerri, Nigeria

Relevant Coursework: Communication Networks, Data Communications, Computer Programming, Microcomputer Systems, Fields and Waves in Communications, Communication Antennas and Measurements, Microwave Communication System Analysis and Design, Principles of Radar and Navigation systems.

ACADEMIC RESEARCH EXPERIENCE

Research Assistant

Aug 2021 – Aug 2025

Rensselaer Polytechnic Institute

U.S National Science Foundation (NSF) Grant: “CNS Core: Small: Next Generation Tiered Spectrum Licensing” (Award No. 2007454)

- Developed deep-learning algorithms for automated spectrum sharing enforcement, including Convolutional Neural Network and Graph Neural Network architectures for detecting and localizing unauthorized transmitters in dynamic spectrum access systems.
- Designed a privacy-preserving technique that protects incumbent user privacy, such as government radar locations, during spectrum enforcement in the United States' Citizens Broadband Radio Service (CBRS) band.

- Investigated adversarial robustness of deep learning-based spectrum enforcement, identifying and characterizing novel beamforming attack vectors.
- Research resulted in three first-author IEEE publications (one journal, two conferences).

NSF POWDER Platform

Jan 2025 – Dec 2025

- Advanced spectrum sharing research, validated on the NSF Platform for Open Wireless Data-driven Experimental Research (POWDER), toward real-world deployable solutions.

INDUSTRY EXPERIENCE

Wi-Fi Intern

May 2024 – Aug 2024

CableLabs, Louisville, CO

- Designed and implemented a machine learning model to predict end-user Quality of Experience (QoE) from Wi-Fi access-point telemetry, enabling proactive network optimization.
- Authored a white paper on ML-driven Wi-Fi performance prediction.
- Analyzed Wi-Fi KPIs (RSSI, PHY rates, retransmissions, airtime utilization) to identify patterns impacting application-level performance.
- Built a testbed dataset using Candela LANforge and Spirent Umetrix to evaluate the impact of congestion and interference on cloud-gaming QoE.

Network Analytics & Automation Intern

Jun 2023 – Aug 2023

AT&T Labs

- Applied machine learning and data analytics (Pandas, Scikit-Learn, SQL, Snowflake) to analyze large-scale network data across fixed wireless and mobile networks.
- Identified key utilization trends to inform 5G network planning and congestion mitigation strategies for nationwide infrastructure.

AI-Driven 6G Network Research Extern

May 2022 – Aug 2022

IBM Research & RPI

- Built a wireless network simulator integrating Unreal Engine and AirSim to research and evaluate next-generation 6G connectivity scenarios.
- Implemented a machine learning clustering algorithm to optimize drone-assisted network coverage and reliability.

Network Provisioning & Support Engineer

Nov 2019 – Dec 2020

ipNX Nigeria Ltd

- Designed, configured, and optimized IP/MPLS service provider networks using OSPF and BGP. Promoted from internship based on performance.
- Troubleshoot L2/L3 networking issues, restoring connectivity and improving reliability for enterprise customers.

PEER-REVIEWED PUBLICATIONS

- [1] **C. Ezemaduka** and A. A. Abouzeid, "Spec-GNN: Spectrum Enforcement Through Graph Neural Networks in Dynamic Spectrum Access Systems," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 2590–2602, 2026.
- [2] **C. Ezemaduka** and A. A. Abouzeid, "A Beamforming Attack on Deep Learning-Based Spectrum Violator Localization," *IEEE Military Communications Conference (MILCOM) 2025 — WS04: 6G and Spectrum Security for Critical Communication (6GSECC)*, Los Angeles, CA, 2025, pp. 1–6.
- [3] **C. Ezemaduka** and A. A. Abouzeid, "Privacy-aware Deep Learning Based Localization of Spectrum Violators," *IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems (MASS)*, Toronto, Canada, 2023, pp. 98–106.

WHITE PAPER

- **C. Ezemaduka** and L. Hervieu, "Leveraging Wi-Fi Network Information to Predict User Experience," 2024 CableLabs Wireless R&D internal document. Contact Marie-Louise ('Lili') Hervieu, Principal Architect, Wireless R&D CableLabs l.hervieu@cablelabs.com, lili.hervieu@gmail.com to request access.

TEACHING EXPERIENCE

Teaching Assistant, Rensselaer Polytechnic Institute

- ECSE 4660/6660 Internetworking of Things (Spring 2026)
- ECSE 4670 Computer Communications Networks (Fall 2022, Fall 2025)
- ECSE 4520 Communication Systems (Fall 2022)
- ECSE 2660 Computer Architecture, Networks, and Operating Systems (Summer 2021, Fall 2025)
- ECSE 2100 Fields and Waves (Spring 2021)
- ENGR 2350 Embedded Control (Spring 2021)
- ECSE 2410 Signals and Systems (Summer 2026)

TECHNICAL SKILLS

Wireless & Networking: Wi-Fi, 5G/6G, RF Planning, Signal Processing, SDRs (NI USRP), NS-3, GNU Radio, IP/MPLS, OSPF, BGP, Routing & Switching, Network Management & Automation.

Machine Learning: CNN, GAN, GNN, RNN, LSTM, LLMs, NLP, SVM, Decision Trees, Diffusion Models, Reinforcement Learning, PyTorch, TensorFlow, Pandas, Scikit-learn, NumPy.

Programming: Python, C++, MATLAB, SQL.

AWARDS, HONORS & SCHOLARSHIPS

- Inducted into Sigma Xi, The Scientific Research Honor Society (by invitation only), 2025
- Selected participant, IEEE ComSoc School Series on 6G Communication and Wireless Technologies, 2023
- Cisco Global Cyber Operations Scholarship, 2018

- Bertelsmann–Udacity Technology Scholarship for Nanodegree Program, 2020
- MTN Foundation Science & Technology Scholarship, 2014–2017
- Agbami Medical & Engineering Professional Scholarship, 2013–2017
- Mobil Scholarship for Engineering Students, 2013–2017
- Total E&P Scholarship for Engineering Students, 2013–2017
- Research Assistantships, Rensselaer Polytechnic Institute (8 semesters, 2021–2025)
- Teaching Assistantships, Rensselaer Polytechnic Institute (5 semesters, 2021–2026)

PROFESSIONAL SERVICE & MEMBERSHIPS

Paper Reviewer:

- IEEE Transactions on Cognitive Communications and Networking (TCCN), 2026
- IEEE/ACM International Symposium on Quality of Service (IWQoS), 2026
- IEEE International Conference on Emerging Techniques in Computational Intelligence (ICETCI), 2024–2026

Professional Memberships:

- Institute of Electrical and Electronics Engineers (IEEE)
- Sigma Xi: The Scientific Research Honor Society
- National Society of Black Engineers (NSBE)

PROFESSIONAL CERTIFICATIONS

- Cisco Certified Network Associate — Cyber Operations (CCNA CyberOps), 2018–2021
- Machine Learning Certificate, Coursera (Stanford University), 2021
- Professional Leadership Series, Archer Center for Student Leadership Development, Rensselaer Polytechnic Institute, 2022

SELECT AI PROJECTS

- **Adversarial Attack on DL-Based Radar Detection in CBRS:** Evaluated vulnerabilities of deep learning-based Citizens Broadband Radio Service radar detection to adversarial ML attacks and assessed their potential to disrupt spectrum sharing operations.
- **AI & Blockchain Automated Incident Reporting (CrimeCanary):** Developed a deep learning and Algorand blockchain-based system for incentivized, automated incident reporting to reduce law enforcement response times.
- **AI Teaching Assistant with RAG and Guardrails:** Developed and deployed a retrieval-augmented AI assistant for an RPI networking course, using ChromaDB and MiniLM embeddings to provide answers grounded in cited course materials. Implemented guardrails that guide students without revealing homework solutions and support for PDF, image, and packet capture inputs.
- **AI Parking Assistant:** Developed an AI agent integrating live 511NY traffic camera feeds with Claude's vision API to detect parking availability and help drivers locate free parking spots in real time.
- **GNN-Based Movie Recommendation System:** Built a recommendation engine using state-of-the-art Graph Neural Network models to enhance content personalization.
- **3D Pose Estimation:** Designed an LSTM+CNN model for estimating 3D human joint positions from video sequences on the Human3.6M dataset.

- **Conditional GAN for Synthetic Face Generation:** Trained a Conditional Generative Adversarial Network on the Celebrity dataset to generate realistic synthetic human faces.
- **Secure Remote Health Monitoring:** Designed a secure IoT health monitoring system leveraging asymmetric and symmetric encryption with data aggregation at edge nodes.

RESEARCH INTERESTS

Wireless Communications and Networking, Artificial Intelligence, Spectrum Sharing & Management, Network Security & Privacy, Machine Learning Security, Mobile Communications

REFERENCES

Dr. Alhussein Abouzeid — Professor & Department Head, Electrical, Computer, and Systems Engineering, Rensselaer Polytechnic Institute | abouzeid@ecse.rpi.edu

Marie-Louise ('Lili') Hervieu — Principal Architect, Wireless R&D, CableLabs | l.hervieu@cablelabs.com